# E-Commerce Sales Analysis – Final Report

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## 1. Title and Objective

**Project:** E-Commerce Sales Analysis (Data Science Internship, Week 1, WeIntern Pvt Ltd).

**Objective:** clean the Online Retail transaction dataset, compute core sales KPIs, find trends and customer patterns, present them in charts and a Power BI dashboard, and recommend actions. The result is **£10.64M revenue from 19,960 orders**, 85% of it from the UK and about a third of it earned in September to November 2011.

## 2. Dataset Overview

| Item | Detail |
|---|---|
| Source | Online Retail.xlsx (UK-based online gift retailer) |
| Raw size | 541,909 rows × 8 columns |
| Period | 1 Dec 2010 to 9 Dec 2011 |
| Columns | InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country |
| Column types | Identifiers: InvoiceNo, StockCode, CustomerID. Numeric: Quantity, UnitPrice. Date: InvoiceDate. Categorical: Description, Country |
| Derived column | Revenue = Quantity × UnitPrice |
| Final cleaned size | 524,878 rows, 19,960 invoices, 4,338 identified customers, 38 countries |

The raw file was only read, never modified. The cleaned data was saved separately as online_retail_cleaned.csv.

## 3. Cleaning Summary

**Missing values (raw data)**

| Column | Missing | Share | Decision |
|---|---|---|---|
| Description | 1,454 | 0.27% | Rows removed: tiny share, and a product-less line cannot be ranked or interpreted |
| CustomerID | 135,080 | 24.93% | Kept for revenue, orders and product analysis; excluded only from customer-level analysis (392,692 rows kept there) |
| All other columns | 0 | 0% | No action |

**Duplicates:** 5,268 fully identical rows were found and removed. Repeated InvoiceNo values were kept on purpose, because one invoice legitimately holds many product lines. **Validation checks and results**

| Check | Finding | Action |
|---|---|---|
| Quantity should not be negative | Minimum was −80,995 (returns and cancellations) | Kept only Quantity > 0 |
| Price should not be zero or negative | Minimum UnitPrice was −11,062.06 | Kept only UnitPrice > 0 |
| Cancelled invoices | Invoice numbers starting with "C" | Excluded from sales |
| Date format | InvoiceDate converted to datetime; range checked | No invalid dates |
| CustomerID type | Stored as float | Converted to nullable integer |

**Net effect:** 541,909 rows became 524,878 sales rows. The 17,031 rows removed (3.1%) were duplicates, missing descriptions, returns, cancellations and zero or negative prices, each with a reason above.

## 4. KPI Summary

| KPI | Value | Meaning |
|---|---|---|
| Total Revenue | £10,642,110.80 | Sum of Quantity × UnitPrice for valid sales (currency assumed GBP) |
| Total Orders | 19,960 | Unique valid invoices after removing cancellations |
| Average Order Value | £533.17 | Total revenue ÷ total orders |
| Total Units Sold | 5,572,420 | Sum of quantities sold |
| Top Revenue Product | DOTCOM POSTAGE (£206K); best real product: Regency Cakestand 3 Tier (£174K) | Highest revenue line, with and without shipping entries |
| Top Product by Units | Paper Craft, Little Birdie (80,995) | Highest quantity sold, all from one order |
| Top Country | United Kingdom (£9.00M, 84.6%) | Largest revenue contribution |
| Top-Selling Category | Not available | The dataset has no Category column (see section 10) |

## 5. Trend Analysis

| Month | Revenue (£) |
|---|---|
| Dec 2010 | 821,453 |
| Jan 2011 | 689,812 |
| Feb 2011 | 522,546 |
| Mar 2011 | 716,215 |
| Apr 2011 | 536,968 |
| May 2011 | 769,297 |

| Month | Revenue (£) |
|---|---|
| Jun 2011 | 760,547 |
| Jul 2011 | 718,076 |
| Aug 2011 | 757,841 |
| Sep 2011 | 1,056,435 |
| Oct 2011 | 1,151,264 |
| Nov 2011 | 1,503,867 |
| Dec 2011 (to 9 Dec) | 637,790 |

- **Seasonal peak:** revenue stays between £0.5M and £0.8M a month until August, then climbs for three months to £1.50M in November, the highest month. September to November together earn £3.7M, about 35% of annual revenue. This suggests preChristmas stocking by gift buyers and wholesalers, so stock and marketing capacity matter most in this window.
- **December 2011 is incomplete:** the data ends on 9 December, so its low figure is not a decline.
- **Weak spots:** February and April are the lowest months and could be tested with promotions.
- **Regional pattern:** the UK earns £9.0M. The next five markets (Netherlands, EIRE, Germany, France, Australia) earn £0.14M to £0.29M each, so growth outside the UK is small but spread widely.

## 6. Customer Insights

- **Repeat customers dominate:** 65.6% of identified customers bought more than once, and they produce 93.1% of identified-customer revenue. This supports retention and loyalty work over broad acquisition.
- **Revenue is concentrated:** the top 20% of customers account for about 75% of revenue. The largest are customers 14646 (£280K), 18102 (£260K) and 17450 (£194K). • **Some top customers are single-order outliers:** customer 16446 (£168K) is almost entirely invoice 581483, and customer 12346 (£77K) is invoice 541431. These two are one-off bulk orders, not loyal high spenders.
- **Tracking gap:** 25% of sales lines have no CustomerID, so customer behaviour is only visible for about three quarters of transactions.

## 7. Visualization Section

The Power BI page "E-Commerce Sales Analysis Dashboard" (dashboard mockup) combines four KPI cards with the three charts below. Suggested screenshot name: dashboard-mockup.png.

| Chart | Type | What it shows | Interpretation and fix |
|---|---|---|---|
| Monthly Revenue Trend | Line | Revenue over time | Clear Q4 build-up and a spike at the right edge from one £168K order. Plot by month rather than by day so it matches the title |
| Top 10 Products by Revenue | Bar | Revenue by product description | Three of the ten bars (DOTCOM POSTAGE, POSTAGE, Manual) are not products, about £362K in total. Filter them out |
| Revenue Share by Country | Pie | Country share of revenue | One slice is 84.6% and the rest are slivers, which the handbook warns against. Use UK vs International, or a top-5 plus "Other" pie, with a bar chart for the full list |

The notebook also saves a revenue distribution histogram (99th percentile view) and a correlation heatmap, both in outputs/charts. Charts need a title, axis labels and a short insight note beside each one in the final PDF.

## 8. Key Findings

1. Revenue reached £10.64M from 19,960 orders, at an average of £533.17 per order.
2. September to November delivers about 35% of revenue, peaking at £1.50M in November.
3. The UK provides 84.6% of revenue, so the business depends heavily on one market.
4. Repeat customers generate 93.1% of identified-customer revenue.
5. The top 20% of customers drive about 75% of revenue.
6. Shipping and adjustment lines (£362K) appear as top products and distort product rankings.
7. Two bulk orders (invoices 581483 and 541431, £246K together) distort the trend, the top-product-by-units result and the customer ranking.
8. A quarter of sales have no CustomerID, which limits customer analysis.

## 9. Recommendations

Observations above are facts from the data. The actions below are suggestions.

1. **Prepare for September to November:** raise stock and campaign capacity ahead of the peak, since a stock-out in these months costs the most.
2. **Invest in retention:** repeat customers earn over 90% of revenue, so loyalty offers and account management for the top customers are likely to pay back faster than broad acquisition.
3. **Reduce UK dependence:** test targeted promotions in the Netherlands, EIRE, Germany and France.
4. **Capture CustomerID at every sale** to enable segmentation and repeat-purchase analysis on all transactions.
5. **Report products and shipping separately,** and verify the two bulk orders against their cancellation invoices before using the totals in decisions.

## 10. Limitations and Assumptions

- **No Category, payment mode or delivery status columns exist**, so Top-Selling Category and delivery-based analysis could not be done. Product descriptions could be grouped into categories as a future step.
- Currency is assumed to be GBP because the retailer is UK-based; the source does not state it.
- Revenue is gross sales value. Returns and cancellations were excluded, not netted off, so true net revenue is lower.
- Cancellation invoices (starting with "C") were removed, but the original orders they reversed (such as 581483) remain in the totals.
- December 2011 covers only 1 to 9 December.
- Average order value is per invoice and is pulled up by wholesale-sized orders.
- Rows with no Description were removed as non-critical (0.27%). Rows with no CustomerID were kept for non-customer analysis.
- Customer insights cover only the 392,692 rows with a CustomerID.